_____ _____

_____ _____

5 UDP

UDP sendto UDP UDP
UDP
ACK UDP
Tab Ctrl+C UDP
PTY shell PTY PTY UDP
stdout PTY shell top vim

PTY session users.json
PBKDF2-HMAC-SHA256 AUTH
AUTH_OK shell IP
UDP
PTY session bash
PROMPT_COMMAND hook HTTP API UDP
session home shell
Streamlit API

1. UDP socket
2. UDP
3.
4. ACK
5. PTY ANSI

6.
7.
8. API Web session

1.

- IP UDP
- UDP
- ID ACK checksum
-
- shell
- PBKDF2-HMAC-SHA256
-
- ACK UDP
- pwd ls ps ip addr
- client_id session
- session
- HTTP API home
- Streamlit HTTP API

2.

- raw terminal
- Ctrl+C Tab
- ANSI
- PTY ping top/vim
- SIGWINCH TIOCSWINSZ PTY
- 8
- API 127.0.0.1 Bearer token home

1.

- PTY
-

2.

C/S UDP UDP socket client_id AUTH PTY shell session shell selectors UDP socket PTY fd HTTP API session

1. AUTH JSON payload
2. users.json PBKDF2 AUTH_OK AUTH_FAIL
3. raw terminal stdin DATA
4. DATA ACK PTY
5. PTY 1200 DATA
6. DATA ACK stdout
7. HEARTBEAT 15 session
8. bash prompt API JSON Streamlit

28 header + payload header

| □□ | □□ | □□ |
|---|---|---|
| magic | 2 | 0x5554 UDP |
| version | 1 | 1 |
| type | 1 | |
| flags | 1 | |
| header_len | 1 | 28 |
| client_id | 4 | ID session |
| seq | 4 | DATA |

| | | |
|---|---|---|
| ack | 4 | ACK |
| payload_len | 2 | |
| rows | 2 | RESIZE |
| cols | 2 | RESIZE |
| checksum | 4 | CRC32 |

- DATA = 0x01
- ACK = 0x02
- HEARTBEAT = 0x03
- RESIZE = 0x04
- CLOSE = 0x05
- AUTH = 0x06 payload JSON
- AUTH_OK = 0x07 PTY session
- AUTH_FAIL = 0x08 payload

- magic version header_len
- UDP payload_len
- checksum
- type
- client_id DATA RESIZE HEARTBEAT CLOSE AUTH shell

UI

Username: Password:
getpass.getpass() raw terminal PTY
SSH Tab Ctrl+C shell

Streamlit UDP session
client_id
10 HTTP API

lab7/

```
├── udp_remote_terminal/
│   ├── __init__.py
│   ├── client.py
│   ├── monitor.py
│   ├── protocol.py
│   ├── reliable.py
│   ├── server.py
│   ├── streamlit_monitor.py
│   ├── users.json
│   └── user_homes/
├── requirements.txt
└── tests/
    ├── test_monitor.py
    ├── test_protocol.py
    ├── test_reliable.py
    └── test_server_auth.py
```

- UDP socket
- users.json password_hash home
- client_id AUTH PTY
- home PTY chdir shell
- selectors UDP socket PTY fd
- sessions client_id key PTY fd shell pid

- session .udpterm_state/session_<id>.cwd .tsv

bash hook

- snapshot_sessions() snapshot_session() list_session_files() dict HTTP API home
- CLOSE PTY selector fd shell SIGHUP session

API

- monitor.py ThreadingHTTPServer /healthz /api/stats /api/sessions /api/sessions/<client_id> /api/sessions/<client_id>/files
- /healthz --monitor-token /api/* Authorization: Bearer <token>
- HTTP ClientSession

• UDP socket
• raw AUTH AUTH_OK AUTH_FAIL
• stdin raw selector
• stdin UDP stdout
• SIGWINCH RESIZE
• finally CLOSE shell

3.

• protocol.py struct.pack/unpack zlib.crc32 checksum AUTH AUTH_OK AUTH_FAIL unpack_packet() None

• reliable.py ReliableEndpoint _send_queue _unacked _expected_seq _receive_buffer get_packets_to_send() get_packets_to_retransmit() on_data()
• server.py json hashlib.pbkdf2_hmac() hmac.compare_digest() pty.fork() shell selectors UDP socket PTY fd TIOCSWINSZ
• client.py tty.setraw() raw selectors stdin UDP socket
• monitor.py http.server JSON API Bearer token 404/403/400/401
• streamlit_monitor.py Streamlit urllib.request API requests pandas

1. 1

```
python3 -m unittest discover -s lab7/tests
```

• /
• AUTH AUTH_OK AUTH_FAIL
• magic checksum payload header
• payload payload
•

• ACK
•
•
• PBKDF2 home
• DATA shell session
• session 10
• ../
• Monitor API token session

```
wuji@localhost:~/home/networklab/lab7$ python3 -m unittest discover -s tests
.............................
----------------------------------------------------------------------
Ran 29 tests in 0.156s

OK
```

2. 2

```
python3 lab7/udp_remote_terminal/server.py \
  --host 0.0.0.0 \
  --port 9000 \
  --user-db lab7/udp_remote_terminal/users.json \
  --home-root lab7/udp_remote_terminal/user_homes

python3 lab7/udp_remote_terminal/client.py --host 127.0.0.1 --port 9000
```

alice / 123456

```
pwd
touch hello.txt
ls
```

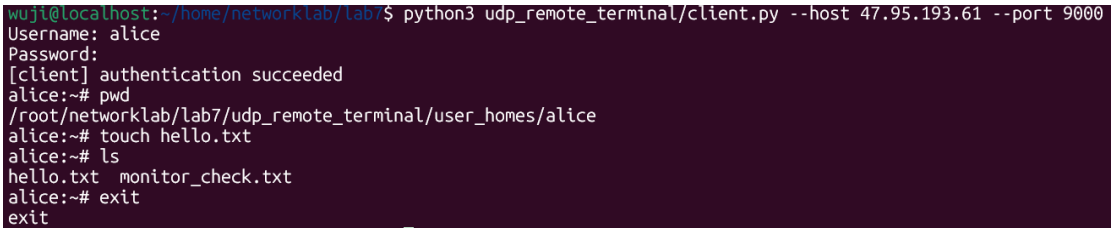
bob / 123456

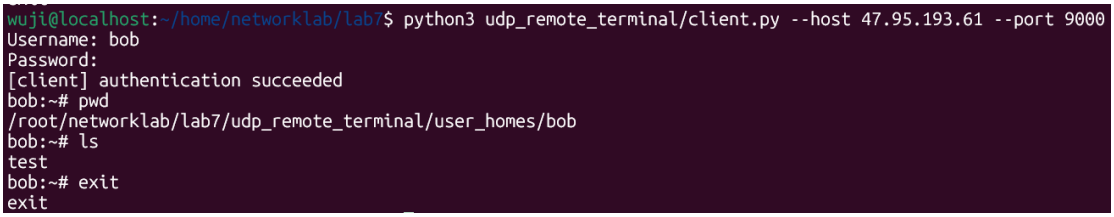
```
pwd
ls
```

alice bob bob alice hello.txt

PTY shell

session AUTH_OK raw terminal PTY




users.json --home-root

3. 3

```
python3 lab7/udp_remote_terminal/server.py \
  --host 0.0.0.0 \
  --port 9000 \
  --user-db lab7/udp_remote_terminal/users.json \
  --home-root lab7/udp_remote_terminal/user_homes
python3 lab7/udp_remote_terminal/client.py --host 127.0.0.1 --port 9000
```

```
pwd
ls
ps
ip addr
```

Linux shell

PTY stdout

```
wuji@localhost:~/home/networklab/lab7$ python3 udp_remote_terminal/client.py --host 47.95.193.61 --port 9000
Username: alice
Password:
[client] authentication succeeded
alice:~# pwd
ls
ps
ip addr
/root/networklab/lab7/udp_remote_terminal/user_homes/alice
hello.txt  monitor_check.txt
    PID TTY          TIME CMD
   1824 pts/1    00:00:00 bash
   1827 pts/1    00:00:00 ps
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:16:3e:53:02:d4 brd ff:ff:ff:ff:ff:ff
    altname enp0s3
    altname ens3
    inet 172.23.184.66/20 metric 100 brd 172.23.191.255 scope global dynamic eth0
       valid_lft 1892159672sec preferred_lft 1892159672sec
    inet6 fe80::216:3eff:fe53:2d4/64 scope link
       valid_lft forever preferred_lft forever
```

4. 4

1.

2.

3. pwd ps

```
root@iZ2zehj1njkff1kf1bwfhdZ:~/networklab# python3 lab7/udp_remote_terminal/server.py   --host 0.0.0.0   --port 9000   --user-db lab7/udp_remote_terminal/users.json   --home-root lab7/udp_re
mote_terminal/user_homes
[server] monitor HTTP API listening on 127.0.0.1:9100
[server] listening on 0.0.0.0:9000, shell=/bin/bash, users=2, home_root=/root/networklab/lab7/udp_remote_terminal/user_homes
[server] client 2810095691 authenticated as 'alice' from ('111.20.225.145', 45977), home=/root/networklab/lab7/udp_remote_terminal/user_homes/alice, shell pid=1911
[server] client 1223298591 authenticated as 'bob' from ('111.20.225.145', 47686), home=/root/networklab/lab7/udp_remote_terminal/user_homes/bob, shell pid=1913
[server] client 1223298591 closed
[server] rejected unauthenticated CLOSE from client 1223298591@('111.20.225.145', 47686)
```

client_id session session PTY fd shell pid

5. 5

1.

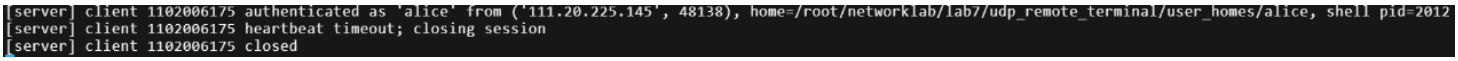
2. session

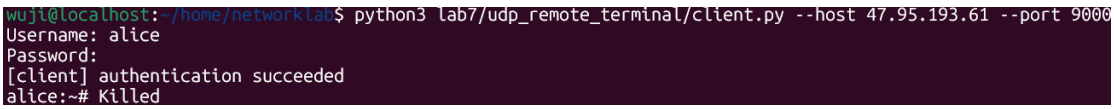
3.

4. --heartbeat-timeout

ACK PTY session

UDP PTY



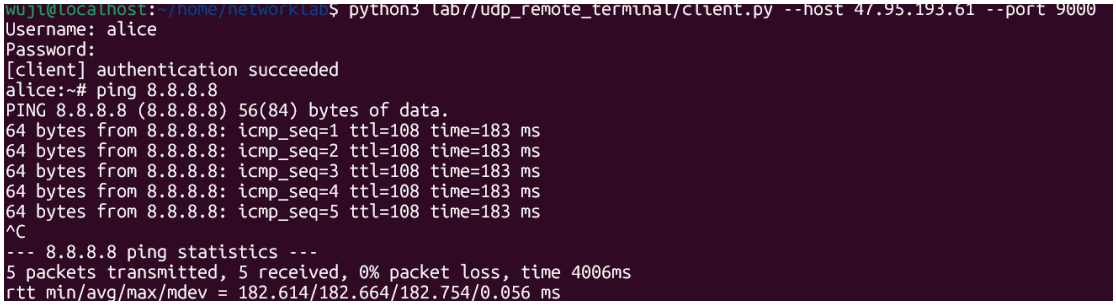




6. 6 Ctrl+C

```
ping 8.8.8.8
```

Ctrl+C

ping Ctrl+C shell




raw Ctrl+C 0x03 PTY PTY

ping

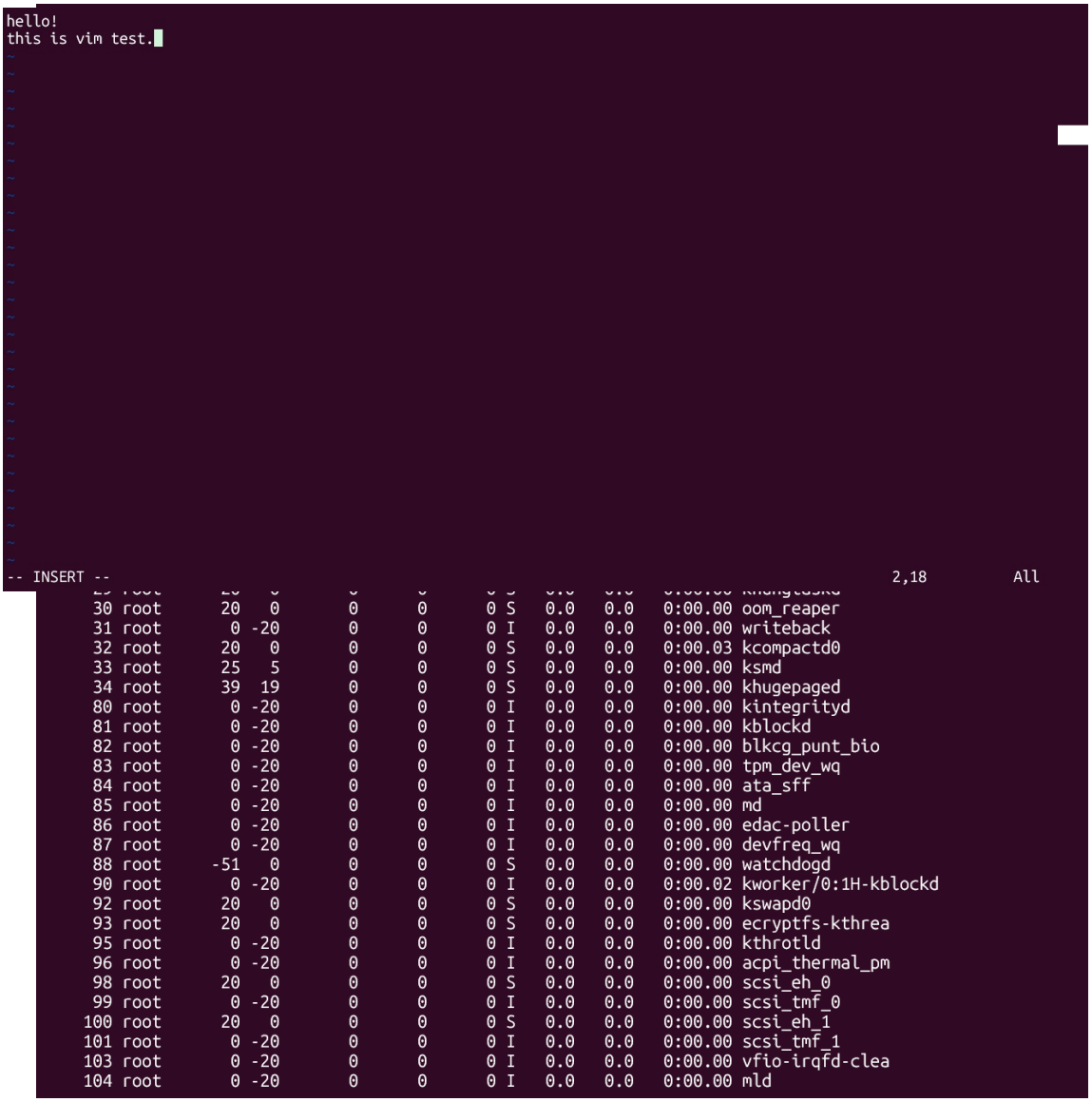
7. 7 top / vim

```
top
```

q

```
vim test.txt
```

top q vim




PTY ANSI ANSI

RESIZE

8. 8 API Streamlit

```
streamlit run lab7/udp_remote_terminal/streamlit_monitor.py \
```

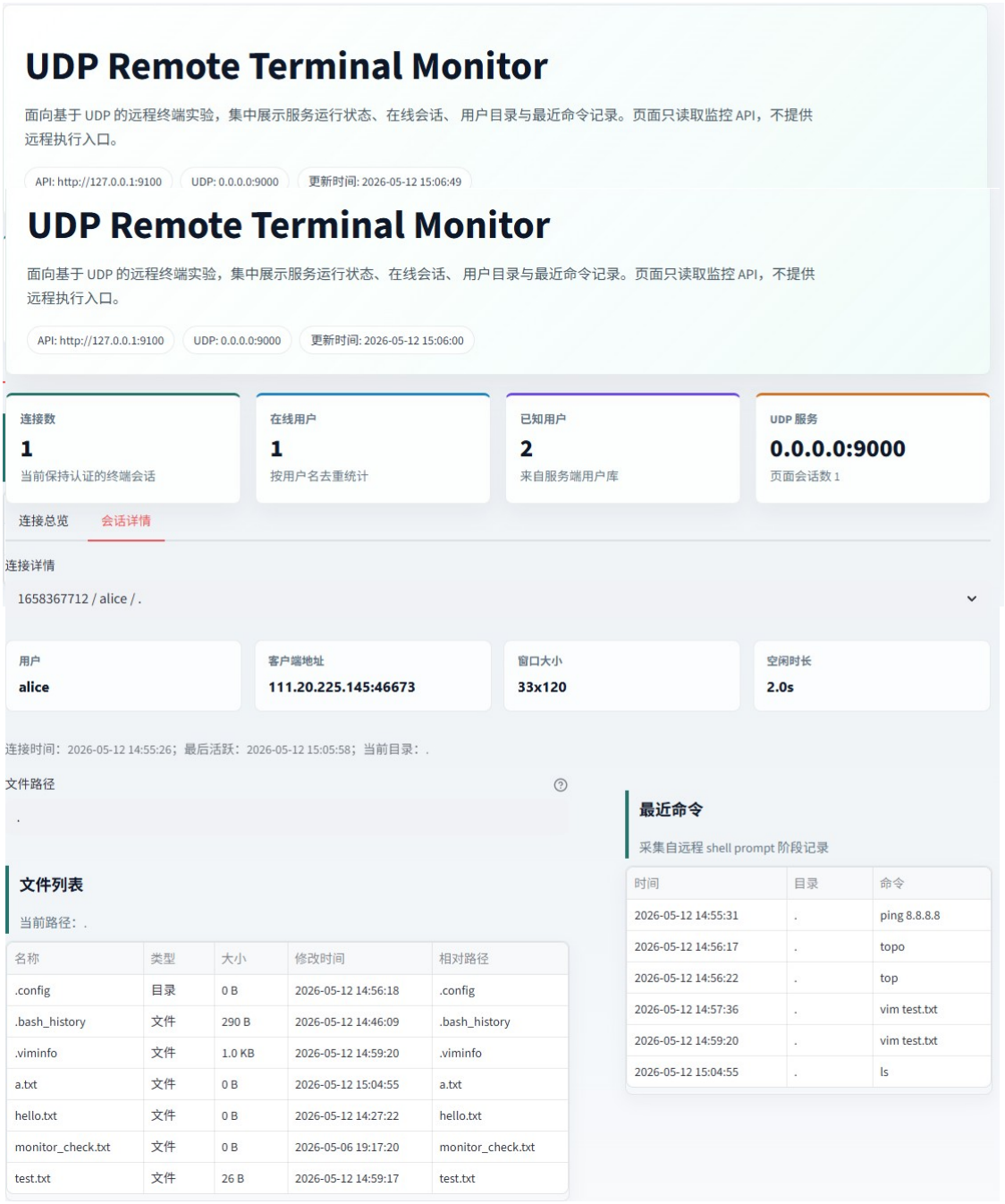
```
-- --api http://127.0.0.1:9100
```

alice

```
pwd
touch a.txt
ls
```

Streamlit alice IP

a.txt pwd touch a.txt ls bob session



PTY shell home

relative_to(home) ../

9. 9

```
sudo tcpdump -i any udp port 9000 -w lab7_udp9000.pcap
```

Wireshark lab7_udp9000.pcap UDP 9000

705 ACK

| | | |
|---|---|---|
| DATA (0x01) | 301 | |
| ACK (0x02) | 348 | |
| HEARTBEAT (0x03) | 33 | |
| RESIZE (0x04) | 14 | |
| CLOSE (0x05) | 2 | |
| AUTH (0x06) | 3 | |
| AUTH_OK (0x07) | 2 | |
| AUTH_FAIL (0x08) | 2 | |

client_id

| **client_id** | | |
|---|---|---|
| 0x0eb7701d | alice | AUTH AUTH_FAIL |
| 0xcdca6556 | alice | |
| 0x1fdcd04c | bob | |

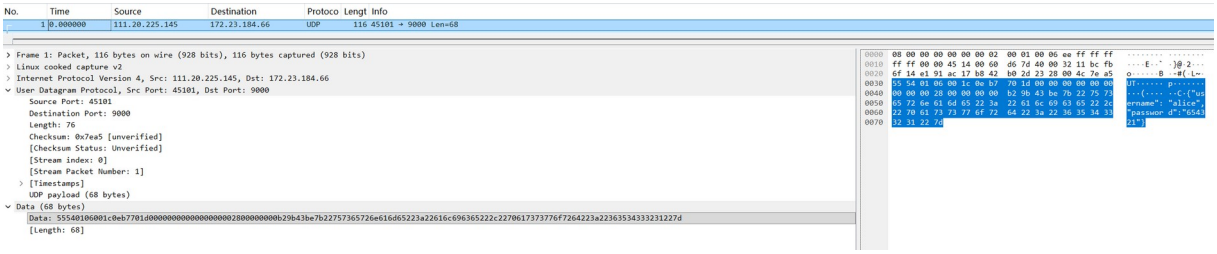
#1 alice AUTH 28 header

```
5554 01 06 00 1c 0eb7701d 00 000 000 00 000 000 0028 0000 0000 b29b43be
```

| | | **Hex** | |
|---|---|---|---|
| 00-01 | magic | 5554 | 0x5554 |
| 02 | version | 01 | 1 |
| 03 | type | 06 | AUTH |
| 04 | flags | 00 | |
| 05 | header_len | 1c | header 28 |
| 06-09 | client_id | 0eb7701d | ID |
| 0a-0d | seq | 00000000 | AUTH 0 |
| 0e-11 | ack | 00000000 | ACK 0 |
| 12-13 | payload_len | 0028 | payload 40 |
| 14-15 | rows | 0000 | AUTH |

| | | | |
|---|---|---|---|
| 16-17 | cols | 0000 | AUTH |
| 18-1b | checksum | b29b43be | CRC32 |

#1 Wireshark UDP payload header JSON payload




payload JSON UDP payload

AUTH AUTH_FAIL

| # | | | client_id | |
|---|---|---|---|---|
| 1 | C→S | AUTH (0x06) | 0x0eb7701d | JSON |
| 2 | S→C | AUTH_FAIL (0x08) | 0x0eb7701d | invalid username or password |

PTY session client_id DATA HEARTBEAT RESIZE

AUTH_OK DATA shell prompt

| # | | | client_id | |
|---|---|---|---|---|
| 3 | C→S | AUTH (0x06) | 0xcdca6556 | JSON |
| 4 | S→C | AUTH_OK (0x07) | 0xcdca6556 | payload ok |
| 5 | S→C | DATA (0x01) | 0xcdca6556 | shell prompt |

PTY session

ls ACK

| # | | | / CK | payload |
|---|---|---|---|---|
| 17 | C→S | DATA | seq=1 | l |
| 18 | S→C | ACK | ack=1 | seq=1 |
| 21 | C→S | DATA | seq=2 | s |
| 22 | S→C | ACK | ack=2 | seq=2 |
| 25 | C→S | DATA | seq=3 | |
| 26 | S→C | ACK | ack=3 | seq=3 |
| 27 | S→C | DATA | seq=5 | shell |
| 28 | S→C | DATA | seq=6 | ls |
| 29 | S→C | DATA | seq=7 | shell prompt |

| | | | | |
|---|---|---|---|---|
| 30-32 | C→S | ACK | ack=5/6/7 | |

DATA ACK

DATA ACK

bob client_id

| # | | | client_id | |
|---|---|---|---|---|
| 385 | C→S | AUTH | 0x1fdcd04c | bob |
| 386 | S→C | AUTH_OK | 0x1fdcd04c | |
| 387 | S→C | DATA | 0x1fdcd04c | bob shell prompt |
| 388 | C→S | RESIZE | 0x1fdcd04c | |
| 389 | C→S | HEARTBEAT | 0x1fdcd04c | |

alice bob client_id client_id session alice

bob

bob cd .. cd ../alice

```
[server] returned to home: access outside ~ is disabled
```

shell prompt home home

chroot Linux

alice 5 HEARTBEAT ACK

| # | | | |
|---|---|---|---|
| 700 | 122.27s | C→S | HEARTBEAT |
| 701 | 122.27s | S→C | ACK |
| 702 | 127.10s | C→S | HEARTBEAT |
| 703 | 127.10s | S→C | ACK |
| 704 | 131.98s | C→S | HEARTBEAT |
| 705 | 131.98s | S→C | ACK |

HEARTBEAT type=0x03 payload_len=0

session heartbeat timeout PTY

session

RESIZE #6 RESIZE header

```
5554 01 04 00 1c cdca6556 00 000 000 00 000 000 0000 0021 0078 66ee0504
```

rows 0x0021 33 cols 0x0078 120

55×210 33×120 SIGWINCH RESIZE

TIOCSWINSZ PTY

bob exit PTY CLOSE CLOSE

| # | | | |
|---|---|---|---|
| 597 | C→S | DATA | bob exit |
| 600 | S→C | DATA | shell exit |
| 601 | S→C | CLOSE | shell |
| 602-603 | C→S | ACK | |
| 604 | C→S | CLOSE | |
| 605 | S→C | AUTH_FAIL | session |

UDP TCP ACK

magic type client_id seq ack payload_len rows cols checksum

UDP UDP ACK UDP PTY raw terminal shell selectors session PBKDF2-HMAC-SHA256 PTY session API Streamlit IP UDP 9000 API 127.0.0.1

home bash PROMPT_COMMAND hook prompt shell home home API relative_to(home) ../ shell Linux PROMPT_COMMAND shell prompt Linux chroot namespace AUTH UDP payload SSH HMAC Linux setuid()/setgid() chroot namespace

UDP socket API

ANSI PTY top vim

TCP

1.

udp_remote_terminal/protocol.py

udp_remote_terminal/reliable.py

udp_remote_terminal/server.py

udp_remote_terminal/client.py

udp_remote_terminal/monitor.py

udp_remote_terminal/streamlit_monitor.py

udp_remote_terminal/users.json

2. tests/test_protocol.py

tests/test_reliable.py

tests/test_server_auth.py

tests/test_monitor.py